

Analysis of microservices resource allocation with elastic resources and requirements

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Context

Modern latency-sensitive applications like *Autonomous Driving Agents* in smart factories and cities generate sensor data requiring real-time processing to respect strict *end-to-end latency constraints* of their applications[1]. Deployment can leverage the entire edge-to-cloud continuum[2] scaling to more costly resources when necessary while ensuring all constraints.

Problem

Deploying such applications via Virtual Network Function Placement Problem (VNFP) sometimes yields infeasible solutions, and the *Placement Models*[6] are unable to create an *Application Deployment Plan* due to constraints. Current models provide no diagnostic insight: critical microservice(s) or binding constraint(s).

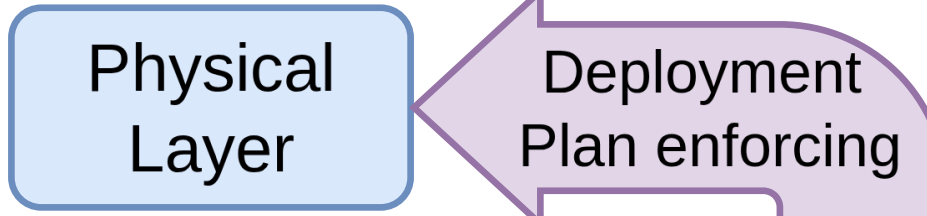
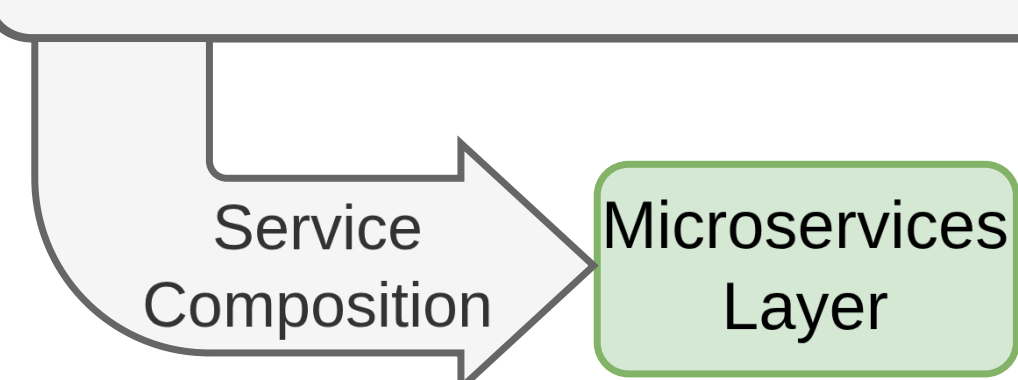
Goal

Develop **elastic optimization models** table to quantify the minimum changes to restore feasibility also identifying primary responsible for infeasibility. Such information act as *feedback* to the *choreographer* in order to support precise diagnosis of deployment failures and to guide corrective actions.

Choreographer

Decentralized approach for *service composition*[3,4].

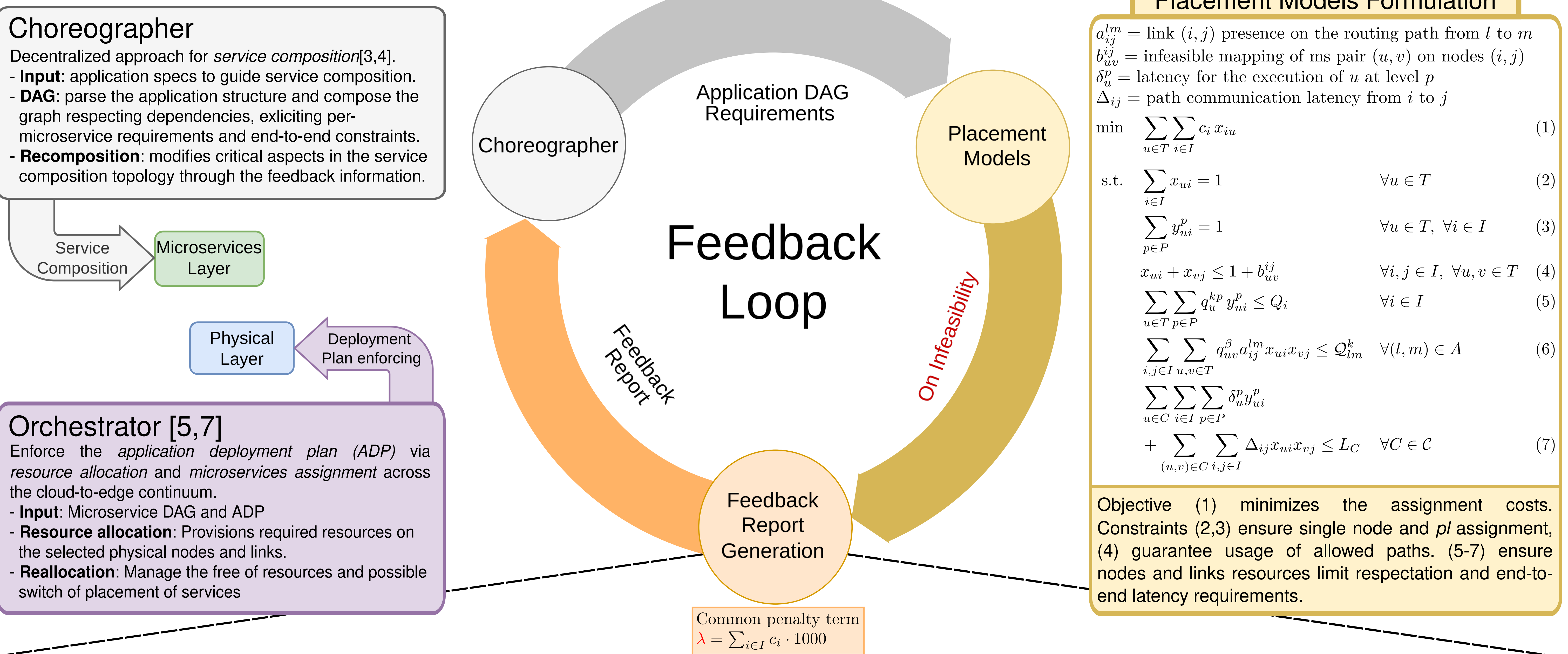
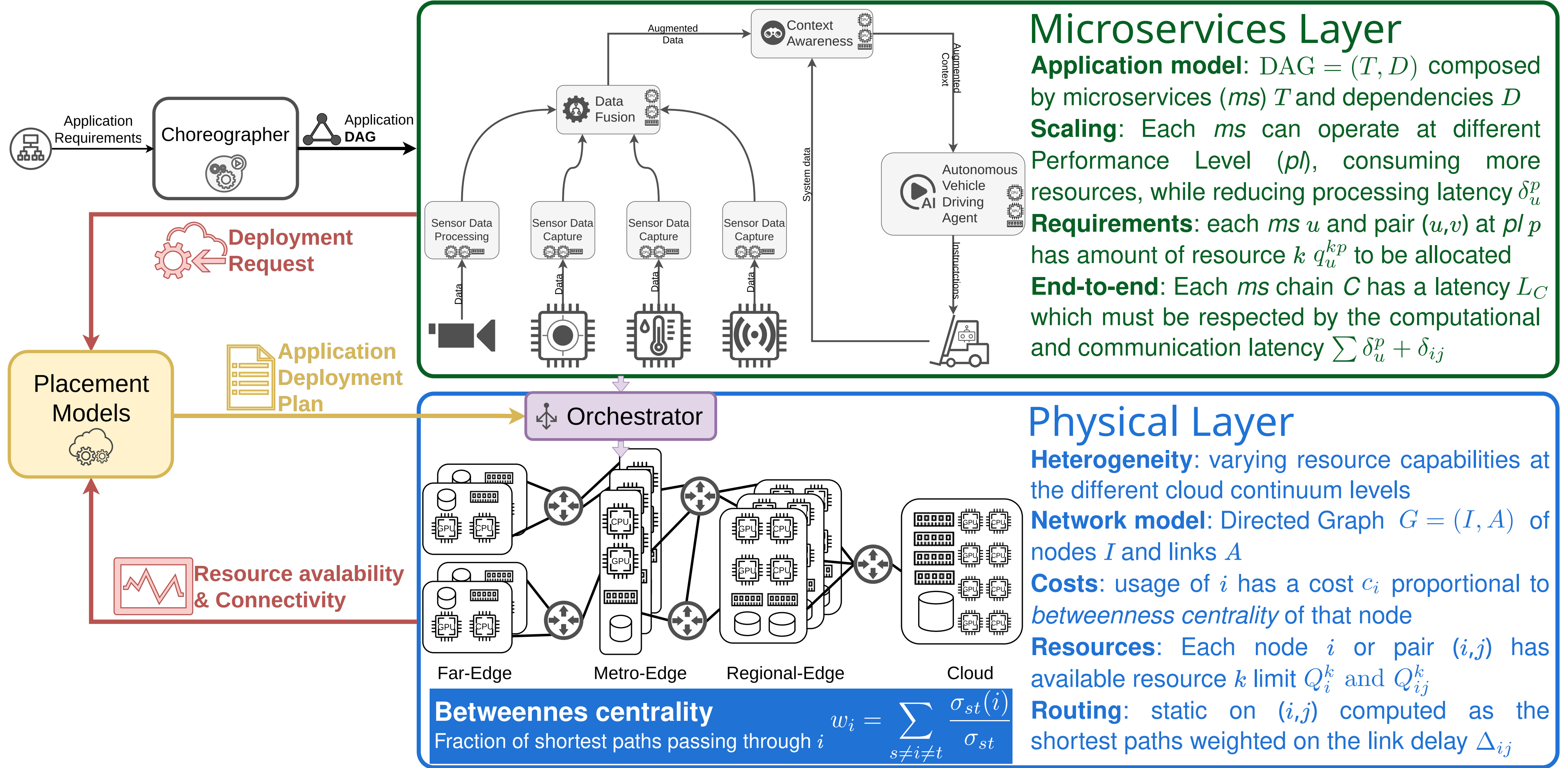
- **Input**: application specs to guide service composition.
- **DAG**: parse the application structure and compose the graph respecting dependencies, eliciting per-microservice requirements and end-to-end constraints.
- **Recomposition**: modifies critical aspects in the service composition topology through the feedback information.



Orchestrator [5,7]

Enforce the *application deployment plan (ADP)* via *resource allocation* and *microservices assignment* across the cloud-to-edge continuum.

- **Input**: Microservice DAG and ADP
- **Resource allocation**: Provisions required resources on the selected physical nodes and links.
- **Reallocation**: Manage the free of resources and possible switch of placement of services



Placement Models Formulation

a_{ij}^{lm} = link (i, j) presence on the routing path from l to m

b_{uv}^{ij} = infeasible mapping of ms pair (u, v) on nodes (i, j)

δ_u^p = latency for the execution of u at level p

Δ_{ij} = path communication latency from i to j

$$\min \sum_{u \in T} \sum_{i \in I} c_i x_{iu} \quad (1)$$

$$\text{s.t.} \quad \sum_{i \in I} x_{ui} = 1 \quad \forall u \in T \quad (2)$$

$$\sum_{p \in P} y_{ui}^p = 1 \quad \forall u \in T, \forall i \in I \quad (3)$$

$$x_{ui} + x_{vj} \leq 1 + b_{uv}^{ij} \quad \forall i, j \in I, \forall u, v \in T \quad (4)$$

$$\sum_{u \in T} \sum_{p \in P} q_u^{kp} y_{ui}^p \leq Q_i^k \quad \forall i \in I \quad (5)$$

$$\sum_{i, j \in I} \sum_{u, v \in T} q_{uv}^{\beta} a_{ij}^{lm} x_{ui} x_{vj} \leq Q_{lm}^k \quad \forall (l, m) \in A \quad (6)$$

$$\sum_{u \in C} \sum_{i \in I} \sum_{p \in P} \delta_u^p y_{ui}^p + \sum_{(u, v) \in C} \sum_{i, j \in I} \Delta_{ij} x_{ui} x_{vj} \leq L_C \quad \forall C \in C \quad (7)$$

Objective (1) minimizes the assignment costs. Constraints (2,3) ensure single node and pl assignment, (4) guarantee usage of allowed paths. (5-7) ensure nodes and links resources limit respectation and end-to-end latency requirements.

Physical Feedback

Microservice Feedback

End-to-End Feedback

Elastic physical node resources

Feedback on: nodes with insufficient resources and quantification of the necessary increase.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{i \in I} \sum_{k \in R} r_i^k \cdot \lambda \quad (8)$$

$$\text{s.t.} \quad \sum_{u \in T} \sum_{p \in P} q_u^{kp} \cdot y_{ui}^p \leq Q_i^k + r_i^k \quad \forall i \in I \quad \forall k \in R \quad (9)$$

$$r_i^k \geq 0 \quad \forall i \in I \quad \forall k \in R \quad (10)$$

(2-4), (6-7)

Low-capacity nodes

Minimal processing requirements shrinking

Feedback on: microservices with excessive minimum requirements and quantification of necessary reduction.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{u \in T} \sum_{i \in I} \sum_{k \in R} r_{ui}^k \cdot \lambda \quad (14)$$

$$\text{s.t.} \quad \sum_{u \in T} \sum_{p \in P} q_u^{kp} \cdot y_{ui}^p - r_{ui}^k \leq Q_i^k \quad \forall i \in I \quad \forall k \in R \quad (14)$$

$$r_{ui}^k \leq M \cdot y_{ui}^0 \quad \forall u \in T \quad \forall i \in I \quad \forall k \in R \quad (16)$$

$$r_{ui}^k \geq 0 \quad \forall u \in T \quad \forall i \in I \quad \forall k \in R \quad (17)$$

(2-4), (6-7)

Oversized microservice requirements

Elastic end-to-end latency requirements

Feedback on: minimal increase of the end-to-end latency bound to make all communication delays feasible.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{c \in C} l_c \cdot \lambda \quad (22)$$

$$\text{s.t.} \quad \sum_{u \in C} \sum_{i \in I} \sum_{p \in P} \delta_u^p y_{ui}^p + \sum_{(u, v) \in C} \sum_{i, j \in I} \Delta_{ij} x_{ui} \cdot x_{vj} \leq L_c + l_c \quad \forall c \in C \quad (23)$$

$$l_c \geq 0 \quad \forall c \in C \quad (24)$$

(2-6)

High latency microservices chain

Elastic physical link resources

Feedback on: links with insufficient resources and quantification of the necessary increase.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{(i, j) \in A} \sum_{k \in R} r_{ik}^k \cdot \lambda \quad (11)$$

$$\text{s.t.} \quad \sum_{i \in I} \sum_{j \in I} \sum_{u \in T} \sum_{v \in T} q_{uv}^k a_{ij}^{lm} x_{ui} \cdot x_{vj} \leq Q_{lm}^k + r_{lm}^k \quad \forall (l, m) \in A \quad \forall k \in R \quad (12)$$

$$r_{lm}^k \geq 0 \quad \forall (l, m) \in A \quad \forall k \in R \quad (13)$$

(2-5), (7)

Low-capacity links

Communication requirements shrinking

Feedback on: communications with excessive minimum requirements and quantification of necessary reduction.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{u, v \in D} \sum_{i, j \in A} \sum_{k \in R} r_{uvij}^k \cdot \lambda \quad (18)$$

$$\text{s.t.} \quad \sum_{i \in I} \sum_{j \in I} \sum_{u \in T} \sum_{v \in T} q_{uv}^k a_{ij}^{lm} x_{ui} \cdot x_{vj} - r_{uvij}^k \leq Q_{lm}^k \quad \forall (l, m) \in A \quad \forall k \in R \quad (19)$$

$$r_{uvij}^k \leq M \cdot y_{ui}^0 y_{vj}^0, \quad r_{uvij}^k \geq 0 \quad \forall u, v \in T \quad (20, 21)$$

(2-5), (7)

High-volume communications

Maximum processing performance discount

Feedback on: minimal resource reduction at maximum pl needed for latency reduction and excess quantification.

$$\min \sum_{u \in T} \sum_{i \in I} c_i \cdot x_{iu} + \sum_{u \in T} \sum_{i \in I} \sum_{k \in R} r_{ui}^k \cdot \lambda \quad (25)$$

$$\text{s.t.} \quad \sum_{u \in T} \sum_{p \in P} q_u^{kp} \cdot y_{ui}^p - r_{ui}^k \leq Q_i^k \quad \forall i \in I \quad \forall k \in R \quad (26)$$

$$r_{ui}^k \leq M \cdot y_{ui}^{\max} \quad \forall u \in T \quad \forall i \in I \quad \forall k \in R \quad (27)$$

$$r_{ui}^k \geq 0 \quad \forall u \in T \quad \forall i \in I \quad \forall k \in R \quad (28)$$

(2-4), (6-7)

Large low processing cost

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